Sean Reiter — Curriculum Vitae

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Education

- **Virginia Tech** — **Blacksburg, VA**
  Ph.D. in Mathematics, Advised by Dr. Mark Embree and Dr. Serkan Gugercin — *May 2025*
  Dissertation: Dimension Reduction in Structured Dynamical Systems: Optimal-$\mathcal{H}_2$ Approximation, Data-Driven Balancing, and Real-Time Monitoring.
  GPA: 3.81/4.00
- **Virginia Tech** — **Blacksburg, VA**
  M.S. in Mathematics, Advised by Dr. Mark Embree and Dr. Serkan Gugercin — *December 2021*
  Thesis: On the Tightness of the Balanced Truncation Error Bound with an Application to Arrowhead Systems
  GPA: 3.81/4.00
- **Virginia Tech** — **Blacksburg, VA**
  B.S. in Mathematics, 'In Honors,' Minor in Chemistry, Summa Cum Laude — *May 2018*
  GPA: 3.86/4.00

Professional Experience

- **Courant Instructor** — **New York, NY**
  Courant Institute of Mathematical Sciences, New York University — *August 2025 - Present*
- **Graduate Research Assistant** — **Blacksburg, VA**
  Department of Mathematics, Virginia Tech — *January 2019 - May 2025*
- **Givens Associate** — **Lemont, IL**
  MCS Division, Argonne National Laboratory — *June - August 2022, 2023*

Research Interests

- Model-order reduction of large-scale dynamical systems, optimal model-order reduction, structure-preserving model-order reduction, data-driven methods for reduced-order modeling and system realization, rational approximation, low-rank matrix decompositions, scientific machine learning, data assimilation, filtering

Publications

Journal Articles

- Reiter, S., and Werner, S. W. R. (2026). Data-driven balanced truncation for second-order systems with generalized proportional damping. *SIAM Journal on Scientific Computing*, 48(3):C526–C552. `10.1137/25M1768217`.
- Reiter, S., Pontes Duff, I., Gosea, I. V., and Gugercin, S. (2026). $\mathcal{H}_2$ Optimal Model Reduction of Linear Systems with Multiple Quadratic Outputs. *IEEE Transactions on Automatic Control*, 71(5):3168–3183. `10.1109/TAC.2025.3636441`.
- Reiter, S., Gosea, I. V., and Gugercin, S. (2025). Generalizations of data-driven balancing: What to sample for different balancing-based reduced models. *Automatica*, 182:112518. `10.1016/j.automatica.2025.112518`.
- Reiter, S., Damm, T., Embree, M., and Gugercin, S. (2024). On the balanced truncation error bound and sign parameters from arrowhead realizations. *Advances in Computational Mathematics*, 50:10. `10.1007/s10444-024-10105-y`.

Conference Proceedings

- Reiter, S., and Werner, S. W. R. (2025). Interpolatory model reduction of dynamical systems with root mean squared error. *IFAC-PapersOnLine*, 59(1):385–390. `10.1016/j.ifacol.2025.03.066`.

Accepted

- Reiter, S., Embree, M., Gugercin, S., and Kekatos, V. (2026). Interpolatory Approximations of PMU Data: Dimension Reduction and Pilot Selection. *IEEE Transactions on Power Systems* (accepted). `10.1109/TPWRS.2026.3707037`.
- Reiter, S., Gosea, I. V., Pontes Duff, I., and Gugercin, S. $\mathcal{H}_2$-optimal model reduction of linear quadratic-output systems by multivariate rational interpolation. *SIAM Journal on Matrix Analysis and Applications* (accepted). *arXiv preprint arXiv:2505.03057*. `10.48550/arXiv.2505.03057`.

Submitted (arXiv preprints)

- Reiter, S., and Werner, S. W. R. (2026). Symmetric Hermite quadrature-based balanced truncation for learning linear dynamical systems from derivative data. *arXiv preprint arXiv:2606.00298*. `10.48550/arXiv.2606.00298`.
- Ackermann, M. S., Reiter, S., and Trefethen, L. N. (2025). L2 and Linfinity rational approximation. *arXiv preprint arXiv:2512.23357*. `10.48550/arXiv.2512.23357`.

Theses

- Reiter, S. J. (2025). *Dimension Reduction in Structured Dynamical Systems: Optimal-H2 Approximation, Data-Driven Balancing, and Real-Time Monitoring*. Ph.D. dissertation, Virginia Polytechnic Institute and State University. `hdl:10919/134223`.
- Reiter, S. J. (2022). *On the Tightness of the Balanced Truncation Error Bound with an Application to Arrowhead Systems*. M.S. thesis, Virginia Polytechnic Institute and State University. `hdl:10919/107999`.

Conference Contributions

Conference and Oral Presentations

- The Loewner Framework Beyond Linear Outputs. *Young Mathematicians in Model Order Reduction (YMMOR) Conference 2026*, May 26 - 29, 2026, Blacksburg, VA, United States
- The Loewner Framework Beyond Linear Outputs. *27th Conference of the International Linear Algebra Society, ILAS 2026*, May 18 - 22, 2026, Blacksburg, VA, United States
- Optimal-$\mathcal{H}_2$ approximation of linear quadratic output systems by multivariate rational interpolation. *Joint Mathematics Meeting 2026*, Jan 4 - 7, 2026, Washington, D.C.
- $\mathcal{H}_2$-optimal model reduction by multivariate rational interpolation. *Challenges, Opportunities, and New Horizons in Rational Approximation*, Apr 6 - 11, 2025, Banff, Canada
- Interpolatory model reduction of dynamical systems with root mean squared error. *MATHMOD 2025, 11th Vienna International Conference on Mathematical Modeling*, Feb 19 - 21, 2025 Vienna, Austria
- Interpolatory $\mathcal{H}_2$ optimal model reduction of linear systems with quadratic outputs. *Model Reduction and Surrogate Modeling (MORe24)*, Sep 9 - 13 , 2024, La Jolla, CA, United States
- Generalizations of data-driven balancing: what to sample for different Riccati equation-based variants. *SIAM Conference on Applied Linear Algebra*, May 13 - 17, 2024, Paris, France
- $\mathcal{H}_2$-optimal model reduction of linear systems with quadratic outputs. *Young Mathematicians in Model Order Reduction (YMMOR) Conference 2024*, March 4 - 8, 2024, Stuttgart, Germany
- Generalizations of data-driven balancing: What do you need to sample for different balancing-based reduced models? *Mid-Atlantic Numerical Analysis Day*, November 10, 2023, Philadelphia, PA, United States
- Interpolatory Matrix Approximations for Pilot Bus Selection and Disturbance Detection. *2023 Algorithms for Threat Detection PI Workshop*, October 10 - 12, 2023, Fairfax, VA, United States
- Interpolation-based $\mathcal{H}_2$-optimal model reduction of systems with quadratic outputs. *Nonlinear Model Reduction for Control*, May 22 - 26, 2023, Blacksburg, VA, United States
- Structure-preserving Extensions of the QuadBT Framework. *SIAM Southeastern Atlantic Section Annual Meeting*,

March 25 - 26, 2023, Blacksburg, VA, United States

- Power System Event Localization via DEIM. *SIAM Conference on Computational Science and Engineering (CSE23)*, February 26 - March 3, 2023, Amsterdam, The Netherlands

Poster Presentations

- Interpolatory Approximations for PMU Data: Dimension Reduction, Pilot Bus Selection, and Event Detection. *2024 AMPS/ATD PI Workshop*, October 7 - 9, 2024, Alexandria, VA, United States
- On Balanced Truncation Error Bound and Sign Parameters. *Model Reduction and Surrogate Modeling (MORe22) 2022*, September 19 - 23, 2022, Berlin, Germany
- On the Tightness of the Balanced Truncation Error Bound, with an Application to Arrowhead Systems. *Southeast Control Conference*, November 29 - 30, 2021, Blacksburg, VA, United States
- A Tight Balanced Truncation Error Bound for a Certain Class of SISO Systems. *SIAM Conference on Computational Science and Engineering*, March 1 - 5, 2021, Virtual

Teaching Experience

- **Instructor of Record** — **New York Univeristy, New York NY**
  MATH-UA 4424: Numerical Analysis — *Spring 2026*
- **Instructor of Record** — **New York Univeristy, New York NY**
  MATH-UA 352: Numerical Analysis — *Fall 2025*
- **Graduate Teaching Assistant** — **Virginia Tech, Blacksburg VA**
  MATH 4864: Computational Modeling and Data Analytics Capstone — *Fall 2024*
- **Graduate Instructor of Record** — **Virginia Tech, Blacksburg VA**
  MATH 2534: Intro to Discrete Math — *Fall 2023*
- **Graduate Instructor of Record** — **Virginia Tech, Blacksburg VA**
  CMDA 1634: Discovering Computational Modeling and Data Analytics — *Fall 2022*
- **Graduate Instructor of Record** — **Virginia Tech, Blacksburg VA**
  MATH 1225: Calculus of a Single Variable — *Fall 2021*
- **Graduate Teaching Assistant** — **Virginia Tech, Blacksburg VA**
  CMDA 1984: First-year Experience — *Fall 2019, 2020*
- **Urban Teachers Resident Teacher** — **New Era Academy, Baltimore MD**
  College and Career Readiness Math — *Fall 2018*

Conference Organization

- Minisymposium organizer "Numerical Linear Algebra Tools for Model Order Reduction", *27th Conference of the International Linear Algebra Society, ILAS 2026*, Blacksburg, VA, United States, May 2026. Co-organizers: Mattia Mannuci (Karlsruhe Institute of Technology)
- Local organizing committee member, *Young Mathematicians in Model Order Reduction (YMMOR) 2026*
- Minisymposium organizer "Recent Advances in Model Order Reduction and Data-driven Modeling", *MATHMOD 2025, 11th Vienna International Conference on Mathematical Modeling*, Vienna, Austria, February 2025. Co-organizers: Hendrik Kleikamp (University of Münster), Steffen W. R. Werner (Virginia Tech), Jens Saak (MPI Magdeburg)
- Organizing committee member, *Young Mathematicians in Model Order Reduction (YMMOR)*

Professional Service and Society Membership

- **Virginia Tech SIAM Student Chapter** — **Virginia Tech, Blacksburg VA**
  Member — *January 2023 - May 2025*
- **Virginia Tech Mathematics Graduate Student Organization** — **Virginia Tech, Blacksburg VA**
  President and Founder — *January 2023 - May 2025*
- **Department of Mathematics Computational Resources Committee** — **Virginia Tech, Blacksburg VA**
  Graduate student representative — *January 2022 - May 2025*
- **Society for Industrial and Applied Mathematics (SIAM)** — **United States**
  Student Member — *January 2021 - May 2025*
- **Department of Mathematics Applied Numerical Analysis Seminar** — **Virginia Tech, Blacksburg VA**
  Organizer — *January 2021 - January 2022*